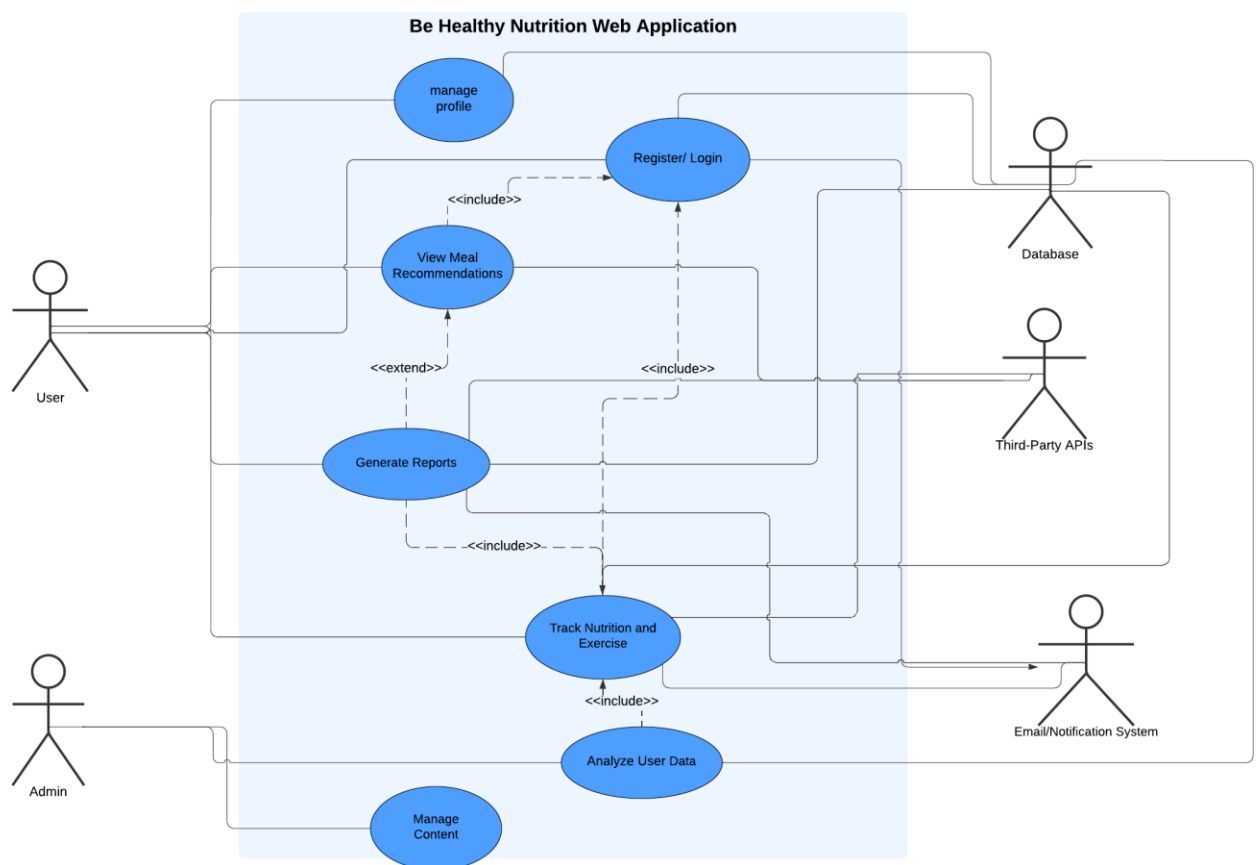


UML use-case diagram for Be Healthy Nutrition Web Application



1. Introduction

The *Be Healthy* project is a nutrition-focused web application aimed at promoting a healthier lifestyle by offering personalized meal recommendations, fitness tracking, and educational resources. This document outlines the **UML Use Case Diagram**, which provides a high-level visualization of system functionality, actors, and relationships.

2. Use Case Diagram Components

2.1. Actors

Actors interact with the system and can be classified as primary or secondary:

Primary Actors

- **User:** Individuals using the platform for meal planning, tracking, and educational resources.
- **Admin:** Platform managers responsible for content management and user data analysis.

Secondary Actors

- **Database:** Stores and retrieves user profiles, meal logs, and fitness data.
 - **Third-Party APIs:** Supplies external data for nutrition and fitness tracking.
 - **Email/Notification System:** Sends notifications and progress updates to users.
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2.2. Use Cases

The key functionalities the system provides are represented as use cases:

- **Register/Login:** Allows users to create accounts and log into the platform.
 - **Manage Profile:** Enables users to update personal details and preferences.
 - **View Meal Recommendations:** Provides tailored meal plans based on user data.
 - **Track Nutrition and Exercise:** Logs user dietary and exercise activities.
 - **Generate Reports:** Creates progress reports for users based on tracked data.
 - **Access Educational Resources:** Offers articles and guides on nutrition and fitness.
 - **Manage Content:** Admin functionality for managing the platform's resources.
 - **Analyze User Data:** Admin functionality for reviewing and analyzing user trends.
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3. Relationships

3.1. Primary Actor Relationships

- **User:**
 - Associated with most user-specific use cases (e.g., Register/Login, View Meal Recommendations).
 - Generate Reports **includes** Track Nutrition and Exercise.
 - Generate Reports **extends** View Meal Recommendations by offering additional insights.
- **Admin:**
 - Associated with admin-only use cases (Manage Content, Analyze User Data).
 - Analyze User Data **includes** Track Nutrition and Exercise for data collection.

3.2. Secondary Actor Relationships

- **Database:**
 - **Association** with Track Nutrition and Exercise, Generate Reports, and Manage Profile.
 - Acts as the central repository for storing and retrieving user and admin data.
- **Third-Party APIs:**
 - **Association** with View Meal Recommendations for external nutritional data.
 - Supports Track Nutrition and Exercise by syncing external fitness tracking data.
- **Email/Notification System:**
 - **Association** with Generate Reports for sending updates.
 - Supports Track Nutrition and Exercise by sending reminders and progress updates.



4. Use Case Diagram Representation

Actors:

Represented as stick figures (e.g., User, Admin, Database).

Use Cases:

Represented as ovals containing the name of the functionality (e.g., Register/Login, View Meal Recommendations).

Relationships:

- **Associations:** Straight lines connecting actors to use cases.
 - **Includes:** Solid arrows pointing from one use case to another with the label «include».
 - **Extends:** Dashed arrows pointing from one use case to another with the label «extend».
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5. Key Insights from the Diagram

- **User-Centric Design:** The majority of functionalities focus on delivering value to the user, such as meal recommendations and progress tracking.
 - **Scalability:** By integrating third-party APIs and a robust database, the system supports scalability and extensibility.
 - **Admin Oversight:** Admins play a critical role in ensuring content quality and analyzing user trends for continuous improvement.
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6. Conclusion

The UML Use Case Diagram for the *Be Healthy* application provides a comprehensive overview of the system's actors, use cases, and relationships. It serves as a blueprint for development, ensuring all functionalities are clearly defined and aligned with user needs. This structured approach will aid in delivering a user-friendly and efficient platform.

